

Naive Bayes: The Mathematics

Contents

1. Overview	2
2. Notation	3
3. Step 1: Bayes' theorem without the evidence	4
4. Step 2: the chain rule	5
5. Step 3: the naive assumption	6
5.1 One factor, checked on the cricket data	7
6. The formula and the MAP rule	8
7. When the assumption fails	10
8. Summary	12
9. Sources	12
10. Key terms	12

Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** [Classification problems](#); [Probability density function \(PDF\)](#); [Independent and mutually exclusive events](#); [Bayes' theorem](#); [Normal distribution](#).

1. Overview

Key point: To score a class, start with how common the class is, then multiply by how often each observed feature value occurs inside that class. The class with the bigger score is the prediction.

The previous Note used the Naive Bayes recipe on a small example. This Note derives it in general, step by step, and makes the “naive” assumption precise.

The recipe first, on the cricket numbers. The new match has toss lost, venue Mumbai and weather sunny. Of the 8 past matches, 5 were wins and 3 were losses. Each score is the **prior** ([the share of all matches that fall in the class, G-1565](#)) times three feature fractions, one factor per line:

Step	Win	Loss
Start: share of matches in the class	$5/8 = 0.625$	$3/8 = 0.375$
× share of the class with toss lost	$\times 1/5 = 0.2$	$\times 2/3 = 0.667$
× share of the class played in Mumbai	$\times 2/5 = 0.4$	$\times 2/3 = 0.667$
× share of the class played in sunny weather	$\times 4/5 = 0.8$	$\times 1/3 = 0.333$
Score after all four factors	0.040	0.056

Loss has the larger score, so the prediction is loss. The rest of the Note shows why this product is the right thing to compute, and when it is not. The scores are later turned into **posteriors** (G-1536; the probability of each class after seeing the features). The formal name for the recipe is the **maximum a posteriori (MAP) rule** (G-1157), and the derivation is summarised in Figure 1.

$$\begin{aligned}
P(C_k | x) &\propto P(x_1, x_2, \dots, x_n, C_k) && \text{Bayes, evidence dropped} \\
&= P(x_1 | x_2, \dots, x_n, C_k) P(x_2 | x_3, \dots, x_n, C_k) \cdots P(x_n | C_k) P(C_k) && \text{chain rule (exact)} \\
&\approx P(x_1 | C_k) P(x_2 | C_k) \cdots P(x_n | C_k) P(C_k) && \text{naive assumption: drop the red parts} \\
&= P(C_k) \prod_{i=1}^n P(x_i | C_k) && \text{the Naive Bayes formula} \\
\hat{y} &= \arg \max_k P(C_k) \prod_{i=1}^n P(x_i | C_k) && \text{maximum a posteriori (MAP) rule}
\end{aligned}$$

Figure 1: The derivation in five lines

2. Notation

Key point: Features $x = (x_1, \dots, x_n)$; K classes, C_1 to C_K for k up to K .

- An **observation** (G-1374; one record, a row of the data table) has n **features** (G-772; input variables, one column each), with values $x = (x_1, x_2, \dots, x_n)$. In the cricket example, $n = 3$: toss, venue and outlook.
- The **target** (G-1949; the output we predict) is one of K classes, $C_1, \dots, C_K$. In the cricket example, $K = 2$: win and loss.

The goal: for each class C_k , compute $P(C_k | x)$, then pick the largest.

	x_1	x_2	x_3	C_k
	toss	venue	outlook	result
new match	lost	Mumbai	sunny	?

features $x = (x_1, x_2, x_3)$, so $n = 3$
target: win or loss
 $K = 2$

Figure 2: The notation on the cricket example: three features and a target with two classes

Figure 2 puts the symbols on the cricket table: each feature column is one x_i , and the result column holds the class.

3. Step 1: Bayes' theorem without the evidence

Key point: $P(C_k | x)$ is proportional to $P(x | C_k) P(C_k) = P(x_1, \dots, x_n, C_k)$.

We want the probability of each class given the features. Bayes' theorem turns that question round, into the probability of the features given the class, which the training data can count:

$$P(C_k | x) = \frac{P(x | C_k) P(C_k)}{P(x)}$$

The **evidence** $P(x)$ (G-718; the probability of the features on their own) is the same for every class, so it does not affect which class is largest (comparing posteriors). Dropping it, we write $\propto$, “is **proportional to**” (G-1584):

$$P(C_k | x) \propto P(x | C_k) P(C_k)$$

Figure 3 checks this on the cricket scores of Section 1. The two scores are 0.040 for win and 0.056 for loss. Dividing both by their sum, the evidence 0.096, gives the posteriors 0.42 and 0.58: the bars change height but not order.

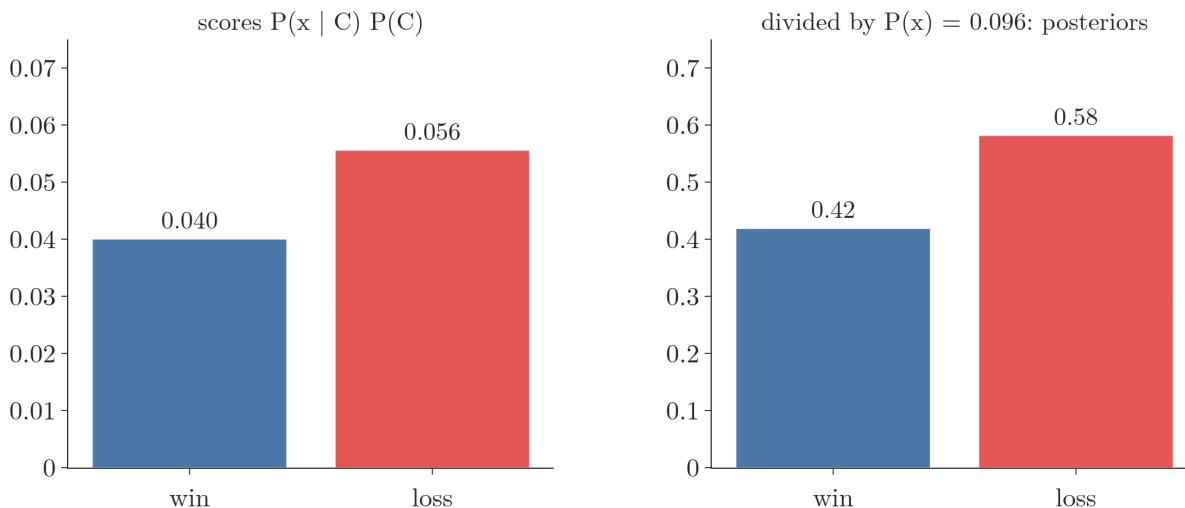


Figure 3: The cricket example. Left: the scores $P(x | C) P(C)$, 0.040 for win and 0.056 for loss. Right: the same scores divided by the evidence $P(x) = 0.096$, the posteriors 0.42 and 0.58. Both pick loss.

From the [proof of Bayes' theorem](#), $P(x | C_k) P(C_k) = P(x \cap C_k)$, the probability of the features and the class together. Writing the features out, with commas meaning “and”:

$$P(C_k | x) \propto P(x_1, x_2, \dots, x_n, C_k)$$

4. Step 2: the chain rule

Key point: Peel off one variable at a time with $P(A, B) = P(A | B) P(B)$. The rule is exact.

The probability of all the features and the class together is hard to count directly. The [multiplication rule of conditional probability](#), $P(A \cap B) = P(A | B) P(B)$, splits it into smaller pieces, one variable at a time. Check it on two events, “win” and “toss lost”. The chance of a win is 5 of the 8 matches, and among the 5 wins the toss was lost once:

$$P(\text{win}) = \frac{5}{8}$$

$$P(\text{toss lost} | \text{win}) = \frac{1}{5}$$

$$\begin{aligned} P(\text{toss lost, win}) &= P(\text{toss lost} | \text{win}) \times P(\text{win}) \\ &= \frac{1}{5} \times \frac{5}{8} \\ &= \frac{1}{8} \end{aligned}$$

Counting directly agrees: 1 of the 8 matches is a win after a lost toss.

Written out for the cricket match. Here $n = 3$, with $x_1 = \text{toss lost}$, $x_2 = \text{Mumbai}$, $x_3 = \text{sunny}$, and $C_k = \text{win}$. Each line peels off one feature:

$$\begin{aligned} &P(x_1, x_2, x_3, \text{win}) \\ &= P(x_1 | x_2, x_3, \text{win}) \times P(x_2, x_3, \text{win}) \end{aligned}$$

$$P(x_2, x_3, \text{win}) = P(x_2 | x_3, \text{win}) \times P(x_3, \text{win})$$

$$P(x_3, \text{win}) = P(x_3 | \text{win}) \times P(\text{win})$$

Substituting the lower lines into the top one gives three conditional factors and the prior:

$$\begin{aligned} &P(x_1, x_2, x_3, \text{win}) \\ &= P(x_1 | x_2, x_3, \text{win}) \\ &\quad \times P(x_2 | x_3, \text{win}) \\ &\quad \times P(x_3 | \text{win}) \end{aligned}$$

$$\times P(\text{win})$$

The same for any n . Apply the rule with $A = x_1$ and $B = (x_2, \dots, x_n, C_k)$. The dots (...) stand for the features in between; for $n = 3$ the list $(x_2, \dots, x_n)$ is just (x_2, x_3) :

$$\begin{aligned} & P(x_1, \dots, x_n, C_k) \\ &= P(x_1 \mid x_2, \dots, x_n, C_k) \\ & \times P(x_2, \dots, x_n, C_k) \end{aligned}$$

The last factor has the same form with one variable fewer, so apply the rule again, with $A = x_2$:

$$\begin{aligned} & P(x_2, \dots, x_n, C_k) \\ &= P(x_2 \mid x_3, \dots, x_n, C_k) \\ & \times P(x_3, \dots, x_n, C_k) \end{aligned}$$

Repeating until only x_n and C_k are left, and finally $P(x_n, C_k) = P(x_n \mid C_k) P(C_k)$:

$$\begin{aligned} & P(x_1, \dots, x_n, C_k) \\ &= P(x_1 \mid x_2, \dots, x_n, C_k) \\ & \times P(x_2 \mid x_3, \dots, x_n, C_k) \\ & \quad \dots \\ & \times P(x_n \mid C_k) \\ & \times P(C_k) \end{aligned}$$

This repeated splitting is the **chain rule of probability** (G-369). Nothing has been assumed yet: the chain rule is exact.

5. Step 3: the naive assumption

Key point: Assume each feature depends only on the class, not on the other features:
 $P(x_i \mid \text{other features}, C_k) = P(x_i \mid C_k)$.

The factors of the chain rule are hard to estimate. For the cricket match, $P(x_1 \mid x_2, x_3, C_k)$ is “the probability the toss was lost, given the venue was Mumbai, the weather was sunny and the match was won”. Few or no training observations match all those conditions, so the estimate is 0 or unreliable ([the problem with specific combinations](#)).

Naive Bayes assumes **conditional independence** (G-443): once the class is known, each feature is independent of the others. In symbols, for any features,

$$P(x_i \mid x_{i+1}, \dots, x_n, C_k) = P(x_i \mid C_k)$$

Conditional independence is the [independence of two events](#), $P(A \mid B) = P(A)$ (knowing B does not change the chance of A), but holding **within each class**. Every chain-rule factor loses its other features (the red parts in Figure 1):

$$\begin{aligned} &P(x_1, \dots, x_n, C_k) \\ &\approx P(x_1 \mid C_k) P(x_2 \mid C_k) \\ &\quad \dots P(x_n \mid C_k) P(C_k) \end{aligned}$$

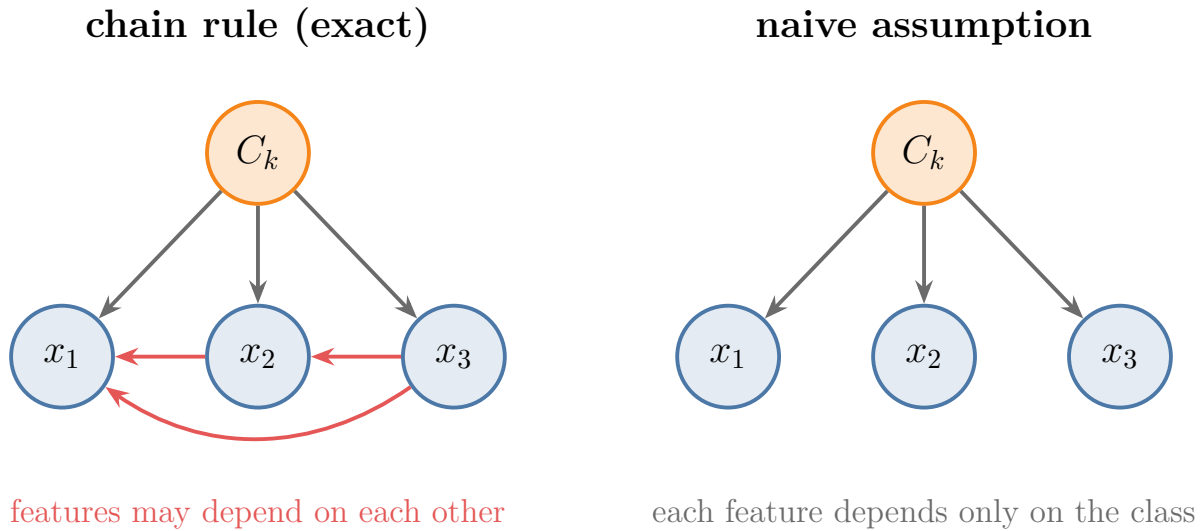


Figure 4: Who depends on whom, for three features. Left: the chain rule allows every feature to depend on the class and on the other features (red arrows). Right: the naive assumption keeps only the arrows from the class

Figure 4 shows what the assumption removes: the red links between features. Each remaining factor needs only one feature and the class, which the training data can count reliably.

5.1 One factor, checked on the cricket data

Key point: The chain-rule factor rests on 1 match; the naive factor rests on 5.

Take the first factor for the class win, with the query of the [cricket example](#) (toss lost, Mumbai, sunny).

1. **The chain-rule factor** is $P(\text{toss lost} \mid \text{Mumbai, sunny, win})$. Only the wins played in Mumbai in sunny weather count. There is 1 such match, and its toss was won. The estimate comes from a single match:

$$0/1 = 0$$

2. **The naive factor** is $P(\text{toss lost} \mid \text{win})$. All 5 wins count, and the toss was lost in 1 of them:

$$1/5 = 0.2$$

Chain-rule factor			Naive factor		
$P(\text{toss lost} \mid \text{Mumbai, sunny, win})$			$P(\text{toss lost} \mid \text{win})$		
<i>toss</i>	<i>venue</i>	<i>outlook</i>	<i>toss</i>	<i>venue</i>	<i>outlook</i>
won	Chennai	sunny	won	Chennai	sunny
won	Mumbai	sunny	won	Mumbai	sunny
lost	Chennai	sunny	lost	Chennai	sunny
won	Mumbai	overcast	won	Mumbai	overcast
won	Chennai	sunny	won	Chennai	sunny
uses 1 winning match toss lost in 0 of 1: estimate 0			uses all 5 winning matches toss lost in 1 of 5: estimate 0.2		

Figure 5: One factor on the five winning matches. Left: the chain-rule factor can use only the win played in Mumbai in sunny weather. Right: the naive factor uses all five wins.

Figure 5 shows the rows each estimate uses. The chain-rule factor is exact in principle, but one match cannot give a reliable estimate, and a 0 wipes out the whole product. The naive factor gives up exactness and gains data: every factor is counted over all the matches of the class.

6. The formula and the MAP rule

Key point: Score each class by $P(C_k) \prod P(x_i \mid C_k)$ and predict the largest: the maximum a posteriori rule.

Written with a product sign. $\prod_{i=1}^n$ means “multiply the terms for $i = 1$, then $i = 2$, up to $i = n$ ”, just as $\sum$ means add. For the cricket match with $n = 3$ and class win:

$$\prod_{i=1}^3 P(x_i \mid \text{win})$$

$$\begin{aligned}
&= P(x_1 \mid \text{win}) \times P(x_2 \mid \text{win}) \\
&\quad \times P(x_3 \mid \text{win}) \\
&= 0.2 \times 0.4 \times 0.8 = 0.064
\end{aligned}$$

$$\begin{aligned}
&P(\text{win}) \times 0.064 \\
&= 0.625 \times 0.064 = 0.040
\end{aligned}$$

The general formula is the same calculation with k and n left open:

$$P(C_k \mid x) \propto P(C_k) \prod_{i=1}^n P(x_i \mid C_k)$$

To get actual probabilities, divide by the sum over all classes. Call that sum Z ; it equals the evidence $P(x)$:

$$P(C_k \mid x) = \frac{1}{Z} P(C_k) \prod_i P(x_i \mid C_k)$$

On the cricket match:

$$Z = 0.040 + 0.056 = 0.096$$

$$P(\text{win} \mid x) = \frac{0.040}{0.096} = 0.42$$

$$P(\text{loss} \mid x) = \frac{0.056}{0.096} = 0.58$$

The prediction is the class with the largest posterior:

$$\hat{y} = \arg \max_{k \in \{1, \dots, K\}} P(C_k) \prod_{i=1}^n P(x_i \mid C_k)$$

($\arg \max$ (G-210) means “the k that gives the maximum”.) This is the **maximum a posteriori (MAP) rule** (G-1157). On the cricket example, it gives 0.040 for win and 0.056 for loss, so $\hat{y} = \text{loss}$.

Figure 6 is a line chart of the two running scores, blue for win and red for loss. The horizontal axis lists the factors in order; the vertical axis is the score so far. The axis is logarithmic: each step up multiplies the value by the same amount, so multiplying by a factor always moves a point down by a distance that depends only on the factor. Each frame multiplies in one more factor, for a match where the toss was lost, the venue was Mumbai and the weather sunny. Win starts ahead on its prior, $5/8$ against $3/8$. The second factor turns it round: only

1 of the 5 wins came after a lost toss, against 2 of the 3 losses. From there on loss stays ahead.

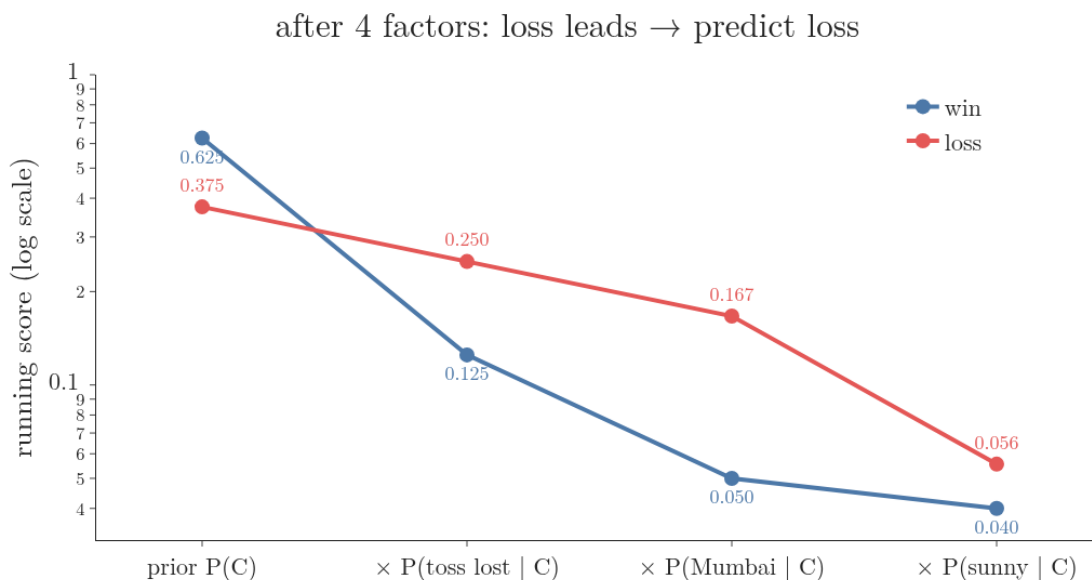


Figure 6: The MAP rule on the cricket example, one factor per frame: the prior, then the probability of each observed feature given the class. Win leads on the prior (0.625 against 0.375) but falls behind after the toss factor; the final scores are 0.040 and 0.056, so the prediction is loss. Log scale.

7. When the assumption fails

Key point: The chain rule is exact; the naive product is only exact when the features really are independent within each class. Strongly related features get counted twice.

A simulation with two binary features (each 0 or 1) and 200,000 observations shows the difference. Each entry is the probability, within class C , that both features equal 1:

Features within a class	Exact $P(x_1, x_2 C)$	Chain rule	Naive product
independent	0.476	0.476	0.477
x_2 is a copy of x_1	0.800	0.800	0.639

When the features are independent, the naive product matches. When x_2 simply repeats x_1 , the naive product multiplies the same evidence twice and is wrong.

An everyday picture: two friends tell us the same rumour, but both read it in the same newspaper. The rumour feels twice as certain, yet we have only one source.

The same information counted again and again makes Naive Bayes over-confident

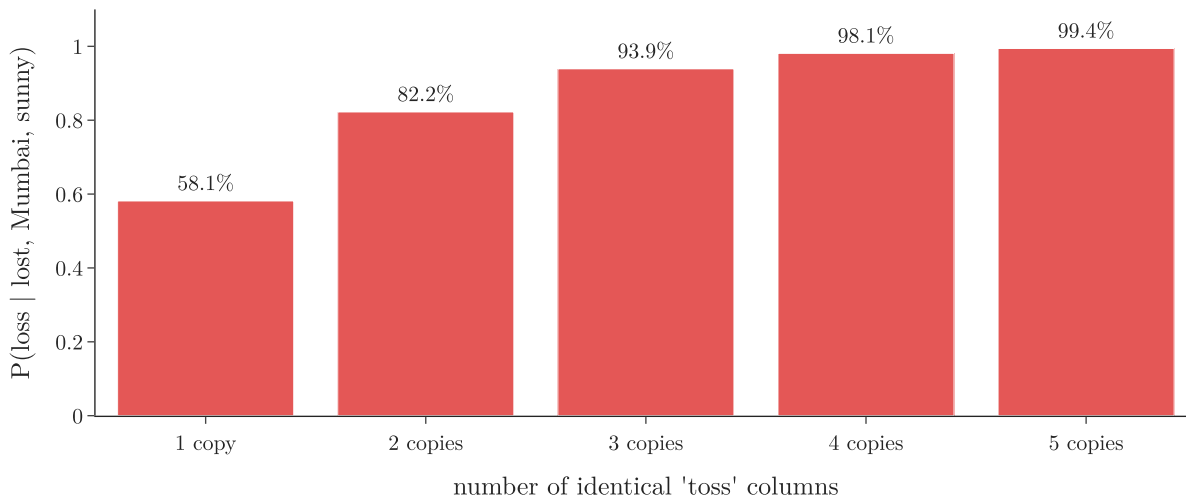


Figure 7: Copying a feature makes Naive Bayes over-confident

Figure 7 is a bar chart. The horizontal axis counts how many identical copies of the “toss” column the model is given; each bar is the probability the model gives to loss (toss lost, Mumbai, sunny). Each extra copy of the “toss” feature pushes $P(\text{loss})$ further, from 58% with one copy to 99.4% with five, although no new information was added.

In practice, Naive Bayes often still picks the right class even when its independence assumption is wrong, because only the **order** of the scores matters for the prediction. Its probabilities, however, tend to be too extreme, so they should not be trusted as exact (Domingos and Pazzani 1997; scikit-learn user guide §1.9).

Extra: Multiplying many probabilities, each below 1, produces very small numbers (the [problem with products](#)). Implementations therefore add logarithms instead:

$$\log P(C_k) + \sum_i \log P(x_i | C_k)$$

The largest sum identifies the same class.

8. Summary

Step	Result
Bayes, drop the evidence	$P(C_k x) \propto P(x_1, \dots, x_n, C_k)$
Chain rule (exact)	$\prod_i P(x_i x_{i+1}, \dots, x_n, C_k) \times P(C_k)$
Naive assumption	$P(x_i \dots, C_k) = P(x_i C_k)$
Formula	$P(C_k) \prod_i P(x_i C_k)$
MAP rule	predict $\arg \max_k$ of the formula

9. Sources

Built from

- CampusX, “Naive Bayes Classifier | Part 7 | Mathematics behind Naive Bayes Algorithm”, YouTube, <https://www.youtube.com/watch?v=2PVRG45eVrY>

Other references

- **Domingos and Pazzani 1997:** Domingos, P. and Pazzani, M. “On the Optimality of the Simple Bayesian Classifier under Zero-One Loss.” *Machine Learning* 29, 103–130, 1997.
- **scikit-learn user guide:** Section 1.9, “Naive Bayes”, scikit-learn 1.9.

10. Key terms

Term	Meaning
Chain rule of probability	Writing a joint probability as a product of conditional probabilities, one variable at a time
Conditional independence	Independence that holds once a third variable (here the class) is known
Proportional to ($\propto$)	Equal up to a constant factor that is the same for every class
$\arg \max$	The value of the variable that makes an expression largest
MAP rule	Maximum a posteriori: predict the class with the largest posterior probability